

Exercise 1.

Exercise 2.

Exercise 3.

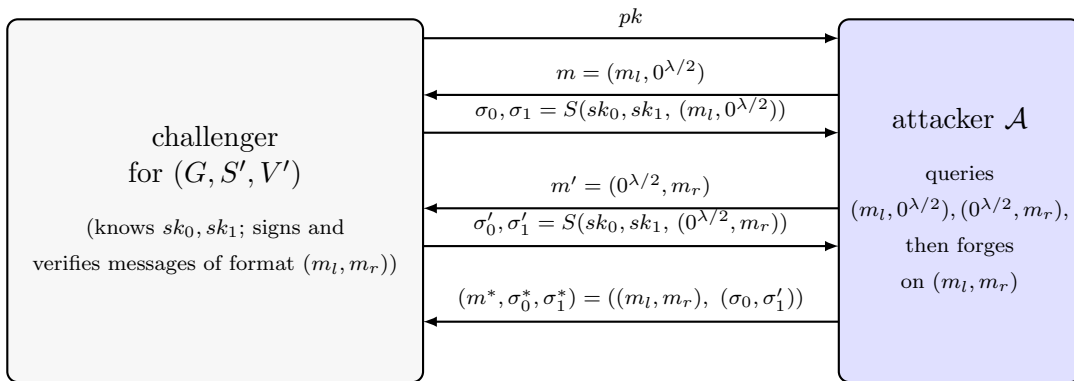
Exercise 4. Let (G, S, V) be a secure signature scheme with the message space $\{0, 1\}^\lambda$. Let $\hat{G}$ be a new key generation function that simply calls G twice and returns a pair of secret keys, (sk_0, sk_1) , and the corresponding pair of public keys, (pk_0, pk_1) . Which of the following signature schemes, $(\hat{G}, \hat{S}, \hat{V})$, are secure? Show an attack or prove security (by reduction).

(a) Prove that “Sign halves” is **not** secure:

$$\hat{S}((sk_0, sk_1), (m_L, m_R)) = (S(sk_0, m_L), S(sk_1, m_R)),$$

$$\hat{V}((pk_0, pk_1), (m_L, m_R), (\sigma_0, \sigma_1)) = V(pk_0, m_L, \sigma_0) \wedge V(pk_1, m_R, \sigma_1).$$

The attack game below shows how the attacker $\mathcal{A}$ can forge a signature for the “sign halves” signature scheme. Let’s discuss the correctness of the attack. First let’s dive



into the freshness of the message. The oracle has only seen the messages $(m_l, 0^{\lambda/2})$ and $(0^{\lambda/2}, m_r)$ and has never been queried for (m_l, m_r) together hence this query does not already exist in its set, being fresh and acceptable. For the case of the signatures being verified, we can check the process. We know that the verification algorithm is $V(pk_0, m_L, \sigma_0) \wedge V(pk_1, m_R, \sigma_1)$. Since both σ'_1 and σ_0 have been queried

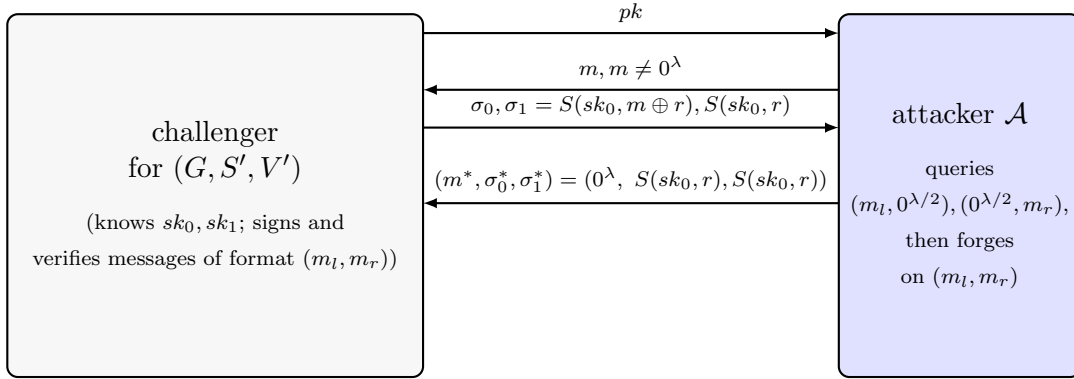
from the oracle they are valid messages for m_r and m_l correspondingly, which means that the verifier will pass.

(b) Show that “Sign with randomness” is **not** secure:

$$\hat{S}(sk_0, m) = [\text{choose random } r \in \{0, 1\}^\lambda; \text{output } (r, S(sk_0, m \oplus r), S(sk_0, r))],$$

$$\hat{V}(pk_0, m, (r, \sigma_0, \sigma_1)) = V(pk_0, m \oplus r, \sigma_0) \wedge V(pk_0, r, \sigma_1).$$

Since the oracle has never been queried for the message zero, this message is fresh



and is accepted. Regarding the verification, the random signature is valid if we put the message as zero for σ_0^* since

$$0^\lambda \oplus r = r$$

.

(c) Prove that “Accept one valid” is secure:

$$\hat{S}((sk_0, sk_1), m) = (S(sk_0, m), S(sk_1, m)),$$

$$\hat{V}((pk_0, pk_1), m, (\sigma_0, \sigma_1)) = V(pk_0, m, \sigma_0) \vee V(pk_1, m, \sigma_1).$$

(Hint: Consider using that $\Pr[\sigma_0 \text{ or } \sigma_1 \text{ is valid}] \leq \Pr[\sigma_0 \text{ valid}] + \Pr[\sigma_1 \text{ valid}]$ from Exercise 4 of last week).